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Inunngorfik – connection to land, inuit culture and giving birth in Kalaallit Nunaat: pregnant women's and their families' perspectives on the centralization of delivery services

Ingelise Olesen[a,b], Christine Ingemann[a,b], Christina Viskum Lytken Larsen[a,b], Charlotte Brandstrup Ottendahl[a], Sofie Emma Rubin[c], Peter Bjerregaard[a] and Tenna Jensen[a,b]

[a]Centre for Public Health in Greenland, National Institute of Public Health, University of Southern Denmark, Copenhagen, Denmark; [b]Greenland Center for Health Research, Institute of Health and Nature, Ilisimatusarfik – University of Greenland, Nuuk, Greenland; [c]Department of Public Health, University of Southern Denmark, Odense, Denmark

ABSTRACT

In Kalaallit Nunaat (Greenland), childbirth is deeply embedded in cultural identity, community, and place. However, the centralization of maternity services increasingly requires pregnant women to travel away from their home communities to give birth, often alone and without familial support. This qualitative study explores the perspectives of 20 pregnant women and their families across the five health regions, using the Peqqissuserput model of Kalaallit Inuit health and wellbeing to understand socio-cultural dimensions of health and well-being. Four key themes emerged: practicalities and birthing, being together or alone, family formation, and identity and sense of belonging. Findings reveal that centralization disrupts Kalaallit Inuit birth culture, diminishes the woman's role as the central figure, and leads to emotional distress, loss of shared family memories, and weakened community ties. In contrast, local births foster emotional support, cultural continuity, and a stronger sense of belongingness to the community for newborns. The study underscores that childbirth is deeply rooted in relational and cultural processes. Drawing on comparative Indigenous models, the findings advocate for culturally grounded, community-based maternity care that restores autonomy, strengthens family and intergenerational bonds, and preserves Inuit birthing practices in Kalaallit Nunaat.

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Introduction

In Kalaallisut (Greenlandic), the moment of a child's birth is expressed as *meeraq inunngorpoq*, which translates to "the child becomes an Inuk (human being)". The phrase reflects a cultural understanding that birth marks not only a biological beginning but also a connection to Inuit culture, community, norms and traditions. Childbirth is a deeply meaningful event in Kalaallit Nunaat (Greenland), symbolising continuity, renewal, and connection to land and community. In this paper, community is understood as the community you are surrounded by in your daily life. The term *maanimioq,* "one who is from here", further emphasises the newborn's rootedness in place, connection to ancestors, belongingness and identity.

The connection to community and the importance of being supported by family has been described in studies conducted among Indigenous people in the Arctic. Research among First Nation communities in Canada shows that local births strengthen women's sense of identity and connection to land and community [1,2]. However, centralisation of birthing services has influenced birthing practices and experiences in Kalaallit Nunaat as well as in other Arctic regions. One of the primary arguments for implementing evacuation policies and centralising birthing services was to reduce high infant mortality rates in remote Arctic regions [3,4]. Since their implementation, these policies have been, and continue to be, debated in terms of their effectiveness and their disconnect to Indigenous birthing practices [4]. More recently, staffing shortages in remote areas have emerged as a central concern, contributing to ongoing challenges in

CONTACT Ingelise Olesen uniiole@uni.gl Centre for Public Health in Greenland, National Institute of Public Health, University of Southern Denmark, Copenhagen, Denmark

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maintaining decentralised maternity care [4,5]. Importantly, this study does not take its point of departure in clinical assessments of risk or medical rationales for centralisation. Instead, it focuses exclusively on understanding what different birthing scenarios mean for the women themselves, their families, and their communities, with particular attention to the psychosocial impact on local culture, family formation patterns, and birthing experiences.

Since the 1970s women in smaller communities in Kalaallit Nunaat have been referred to local health centres for delivery [6], and since 2002, one of the five regional hospitals [7]. This shift was driven by midwife shortages, cost concerns, and efforts to reduce infant mortality [6,8]. The national perinatal guidelines introduced in 2002 further recommended that all complicated births occur at the national hospital in the capital Nuuk and outlined strict criteria for local deliveries [7]. As a result, women from communities without birthing facilities have had to travel by boat or air to Nuuk or the nearest regional hospital to give birth. These pregnant women consequently leave their local communities, usually three weeks before the due date and in some cases even before, depending on the distance. The women are offered accommodation in the regional hospitals' patient hotels. It is not possible for a partner, children, or another companion to stay at the patient hotel.

Currently, births can take place at the national hospital in Nuuk, the health centre in Tasiilaq, and two of the four regional hospitals. Since 2012, facilities in Maniitsoq, Uummannaq, Aasiaat, and Qaqortoq have lost or intermittently lost their birthing status. While infant mortality has declined in Kalaallit Nunaat, studies highlight persistent regional disparities and a developmental gap between Kalaallit Nunaat and Denmark [9]. Figure 1 provides an overview of current and former birthing facilities.

Several studies have explored the broader impacts of centralisation on childbirth, Indigenous women, families, and local communities [3,10–12]. Relocating pregnant women away from their homes and support networks for childbirth has significant consequences. Research shows that women often experience loneliness, and stress when giving birth outside their communities. Nevertheless, many accept the situation out of a desire to protect their unborn child [2,13–15]. In some cases, the lack of support during referred births has led to traumatic experiences, with some women even concealing their pregnancies to avoid relocation. Women also report a loss of autonomy and control over their bodies, as well as economic, emotional, and behavioural impacts on the children left behind [3,15,16]. Therefore, it is important to investigate community perspectives in Kalaallit Nunaat to better understand the nuances of the impact that the centralisation of birthing facilities has, as each context is unique. Our focus is on the human aspects of centralisation, not on infant mortality rates, as these are shaped by a wide range of factors and cannot be definitively compared across Arctic regions.

Aim

The international public health literature suggests that the organisation of childbirth and perinatal care for Indigenous populations are predominantly shaped by biomedical frameworks, often overlooking cultural values and community-based practices. Despite extensive reporting on perinatal outcomes, little is known about pregnant women's preferences and experiences regarding place of birth in Kalaallit Nunaat. National surveys show increasing support for local births—from 74.2% in 1993–94 to 85.3% in 2005–08 [17]. Therefore, this study aims to explore pregnant women and families' perspectives on childbirth in Kalaallit Nunaat in the context of centralised delivery services by addressing the research questions: (1) How do pregnant women and their families perceive the birthing experience? (2) What impact does the centralisation of delivery services have on Kalaallit Inuit birth culture and the understanding of the birthing experience?

Design and method

This study was conducted as part of a research-based consultancy project commissioned by the Ministry of Health in Kalaallit Nunaat. This paper presents selected findings from the project, which also have been published in an alternative format in the full project report, available in Kalaallisut and Danish [5]. Besides the interviews with women and family/community members present in this paper, the project also included interviews with a total of 14 professionals: doctors, midwives, public health nurses, and social workers.

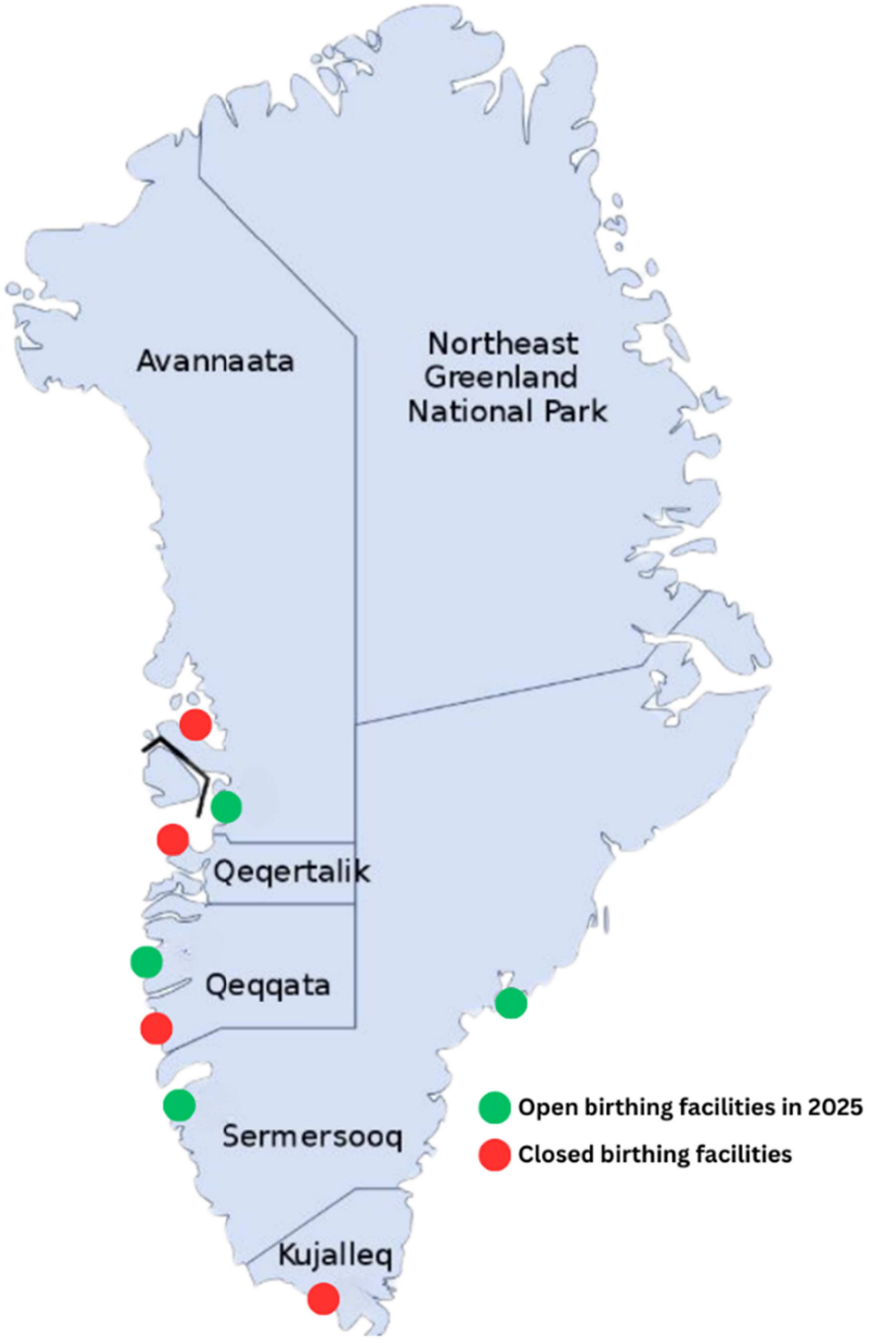


Figure 1. Map of current and former birthing facilities.

Data collection

The study was conducted in the fall of 2022 and the spring of 2023. Qualitative interviews with a semi-structured interview guide [18] were conducted either face-to-face or on the telephone. All interviews were conducted by the first author, who is Kalaallit Inuit and speaks Kalaallisut. She has a background as a midwife and has worked with public health research for more than more than 25 years.

The interview guide was developed using the theoretical research framework Peqqissuserput [19]. The interview guide operationalized the model by including questions that reflect its core determinants. These covered topics such as family relationships, intergenerational perspectives on birthplaces, and previous birth experiences. Participants were also asked about pregnancy, childbirth, and relational dynamics, as well as values

and local strengths. These themes align either directly or indirectly with the Peqqissuserput determinants, e.g. community, family and relationships, culture and language, access to health services, quality of life, and local values. The themes in the interview guide were: 1) General thoughts about places of birth; 2) Experiences with births, own or others' (e.g. giving birth in local community, experiences referring to a region hospital to give birth); 3) Current pregnancy (feeling of security and concerns); 4) Families and relationships; 5) Cultural values and strengths.

Theoretical research framework Peqqissuserput

Peqqissuserput is grounded in Kalaallit Inuit knowledge systems and draws on Indigenous ways of knowing and doing to understand and support health and well-being in communities. The model integrates these knowledge traditions with western public health theory, resulting in a model that highlights the socio-cultural, relational, and contextual dimensions that shape community and health in Kalaallit Nunaat [19].

Central to the model is the concept of *peqqinneq*, which refers to being in the right place—a state of balance in life and fitting into the relationships between land, people, animals and wider forces. In its original meaning, *peqqinneq* also expresses humility and acceptance of human powerlessness in relation to forces greater than oneself. These perspectives position health as a collective, place-embedded experience rather than an individual condition.

This aligns with recent studies indicating that a holistic understanding of life and health remain predominant in present-day Kalaallit Nunaat [20,21]. The Peqqissuserput model therefore illustrates the multifaceted nature of health perceptions in the population and underscores the importance of incorporating these dimensions into health-related research to support communities and families in continuing to thrive.

Interview participants

A total of 20 individuals were interviewed in the study, representing a broad range of experiences with childbirth in Kalaallit Nunaat. The participants included pregnant women, an expecting father, and elderly community members, each offering perspectives shaped by whether births occurred locally or required travel outside the community.

Of the 20 participants, 14 were pregnant women—4 primiparous and 10 multiparous. Their experiences reflected substantial variation in relation to place of birth: some multiparous women had previously given birth locally but were referred during their current pregnancy, while others had only experienced referral-based births. All pregnant women included in the study were experiencing a normal pregnancy without known medical conditions and were expected to have an uncomplicated delivery; women with anticipated complications were excluded, as they would in any case be transferred to the national hospital.

One expecting male participated as well; he was preparing to become a first-time father, and his partner was being referred.

Five elderly community members were interviewed—two women and three men. Each had children of their own and were also grandparents, and all of their children were born locally.

Most participants were recruited with the assistance of midwives working at regional hospitals, and all five health regions in Kalaallit Nunaat were represented. This diversity in geography, experience, and generational background enabled a nuanced understanding of how place, mobility, and community context shape birthing experiences.

All interviews were audio recorded and transcribed. In Kalaallit Nunaat the majority population are Kalaallit Inuit, Indigenous to Kalaallit Nunaat, with Kalaallisut as their primary language. A minority has Danish as their primary language. If interviews were conducted in Kalaallisut they were translated into Danish by the same researcher who collected and later analysed the qualitative data, since the other members of the research team do not speak Kalaallisut. The deductive analysis was based on predetermined themes from the interview guide, and all interviews were coded using NVivo (v.12). During the process, the analysis was further enriched by themes that emerged organically from the interviews, extending beyond the initial framework and allowing for the creation of new thematic connections.

Ethical considerations

The study was performed in accordance with the 2024 Declaration of Helsinki, and the Greenlandic Science Ethics Committee (Danish: Videnskabsetisk Udvalg) granted ethical approval. All participants received both oral

and written information about the study, including participant and layperson information sheets, and they provided written informed consent. The study aligns with the national research strategy, which emphasises that research should be socially relevant and beneficial to Greenlandic society [22]. In this case, relevance and benefit were established through the commissioning of the study by the Ministry of Health in Kalaallit Nunaat, following political concern about the closure of birthing facilities. The final report was subsequently shared and discussed in Inatsisartut (The Parliament of Kalaallit Nunaat). Relevance and legitimacy were further strengthened through a community-based participatory research approach by the involvement of a reference group comprising representatives from the Chief Medical Officer's Office, national health leadership, obstetric leadership, the Midwives' Association, midwives, health visitors, social workers, and the Nursing Ethics Council. The group followed the study throughout and contributed to decisions on study design, participant selection, field sites, and the formulation of the study's development perspectives [23].

Participation was voluntary, and no financial compensation was provided. The interviews focused on taking a strengths-based approach and were conducted by a Kalaallit Inuit midwife and researcher with clinical and cultural expertise in childbirth. This helped minimise potential discomfort when discussing previous birth experiences or concerns about childbirth. The interviewer ensured participants could speak their preferred language and created a supportive environment consistent with a strengths-based and culturally grounded approach. No psychological risks beyond those inherent to qualitative interviews were anticipated, and participants were free to decline or pause any question.

Results

In the exploration of pregnant women and their families' perspectives on childbirth in Kalaallit Nunaat in the context of centralised delivery services, four overarching themes were identified: (1) Practicalities and birthing, (2) Being together or alone, (3) Family formation, and (4) Identity and sense of belonging. The results are also summarised and presented in Table 1.

Practicalities and birthing

Logistical challenges often dominated the thoughts and expectations surrounding pregnancy and childbirth of the women who knew they would be giving birth outside their local community. Preferences for

Table 1. Overview of the factors influencing women's birthing experience.

	Women giving birth at a referral hospital	Women giving birth at their local hospital
Practicalities and birthing	• Concerns about logistics, accommodation, and conditions in connection to travelling.	• Able to concentrate only on the pregnancy and delivery itself. • Familiarity with the place of birth and the local midwives.
Being together or alone	• Experience of loneliness and worries about having to travel and give birth alone. • Concern about not being able to have the partner present.	• Being surrounded by a supportive family during labour. • The family and community can soon after delivery visit and see that the mother and newborn are doing well.
Family formation	• Using available communication technologies to allow partners, siblings, and other family members to participate in the childbirth. • Family members struggle to fully connect through a video call from afar.	• Family event: partner, other children, and family present at birth. • Community event. • Experience of postpartum calm/rest. • Family support in establishing breastfeeding.
Identity and sense of belonging	• Irrespective of local preparedness measures, participants underscored the significance and benefits of community birth and described a sense of loss when births took place in referred hospital settings.	• The community's joy of welcoming a newborn to their community.

which regional hospital to give birth in were largely influenced by whether the women had family or relatives nearby, as well as practical considerations such as travel conditions and financial concerns related to whether they could afford being accompanied by their partner or another accompanying person. One woman shared how she and her partner made the decision to use their own savings so they could be together for the birth:

> *'We rented an apartment at our own expense. It was both a practical and financial challenge. It is very expensive for our family.'*

Some pregnant women were fortunate to have relatives nearby the place they were to give birth, where the partner could stay, or they had the financial means to cover travel expenses. Others did not have that option. The cost of travel (flights and accommodation) could force partners to remain at home, as it became too expensive for the family to afford. The costs of bringing a partner or relative to the place of birth thereby results in social and cultural inequalities in relation to pregnancy and childbirth as income becomes a direct prerequisite for being able to experience the birth as a family.

These findings underscore how the location of birth, whether at their local hospital or away, significantly shapes the practical and emotional landscape of pregnancy, with logistical and financial burdens disproportionately affecting those who must travel for childbirth.

The advantages when giving birth at your local hospital

Giving birth at one's local hospital, surrounded by family, was described by the women as a very different experience. Pregnant women who could give birth at their local hospital were more focused on the birth itself. Their thoughts revolved around practical matters, such as arranging childcare for their other children while they were in the maternity ward. These families were familiar with the hospital staff and midwives. This familiarity brought a sense of comfort and continuity, which was absent when giving birth far from home. A woman explained how she appreciated meeting different midwives during her pregnancy:

> *'I thought it was good that I met different midwives, because then I could relate to the one on duty when I had to give birth.'*

For pregnant women expecting to give birth in their local community, there was a strong focus on the timing of labour. They often expressed specific wishes and expectations for the upcoming birth. Many had made detailed plans and thought carefully about what to do when labour begins, reflecting a sense of preparedness and connection to their familiar surroundings. Giving birth in one's local community, surrounded by family, offered the new mother the opportunity to rest and focus on establishing breastfeeding:

> *'I hope there's room for me to stay at the hospital for a few days after having my child, just until breastfeeding is well established.'*

These findings are also shaped by birth location, it highlights how local births foster a sense of relational continuity and emotional security; elements deeply embedded in cultural expectations around care, family presence, and preparedness for motherhood.

Being together or alone

Women who gave birth at their local hospital described a sense of rest and relaxation. This was largely because family members, relatives, and friends had already visited the maternity ward and met the newborn. Everyone had seen that both mother and baby were healthy and doing well, which brought a sense of calm and reassurance. A mother shared how her family waited outside the delivery room for the newborn to arrive:

> *'I have a huge family. They were all waiting at the hospital when I was in labour (...). They stood outside the delivery room, and as soon as our child was born, they all came in to say hello.'*

Another mother shared how having her family visit her at the hospital made it possible for her to have a peaceful beginning when returning home:

'Family and relatives, who wanted to see the newborn, could visit the hospital. That way, all the maternity visits were done before going home and you could return home in peace, without visitors.'

Being surrounded by a supportive family during labour was deeply important. According to the pregnant women, having family support both at the onset of labour and throughout the birthing process made a significant difference:

'It was great that my mother and my family were with me when I gave birth. My mother could see I needed rest and told me to try to sleep. It helped me a lot to have their support during labour.'

The disadvantages when giving birth alone

One of the most significant challenges for pregnant women referred to give birth at the regional hospitals was having to leave behind their partners, children, and families. Their focus often shifted away from the pregnancy and upcoming birth to the emotional strain of having to travel alone to deliver in an unfamiliar setting, while being separated from their family and community. For many women in Kalaallit Nunaat, giving birth without the presence of a partner or close relatives is not a choice, but a reality. A pregnant woman described her reaction upon learning she would have to give birth away from her community:

'Being alone [during childbirth]. How should I describe it? It felt empty. I felt powerless and alone.'

The thought of facing childbirth alone was overwhelming. It drained her energy and stripped away her sense of strength and empowerment:

'I don't know how many times I cried. I had the feeling that my partner couldn't come along, couldn't be there with me.'

All the pregnant women expressed a strong desire to give birth in their local communities, primarily so they could be close to their partners, children, and extended families. One woman, referred to the regional hospital for her current pregnancy, reflected on the stark contrast between the birth of her first child, which took place in her home community surrounded by a large, supportive family, and now having to face childbirth alone. She described an overwhelming sense of loneliness that deeply affects her emotional well-being:

'When I gave birth for the first time in [home community], my entire family was gathered in our home. Now, giving birth here in Nuuk, I'll go into labour alone, walk to the maternity ward alone, and when I leave, I'll travel home alone with my newborn.'

Travelling alone to give birth was described not only as an experience of deep loneliness, but also as a source of practical concern. One pregnant woman, for example, expressed how she worried about managing the journey back to her local community with a newborn without any support:

'I was worried about travelling home alone with my newborn. I kept thinking: what if I make a mistake or something goes wrong? It's almost autumn, and I imagined slipping and falling. I started imagining all kinds of accidents while handling the baby.'

Feelings of loneliness and lack of support left the women emotionally vulnerable, giving rise to many worries and thoughts. These were concerns they might not have experienced under different circumstances, such as having their partner or an accompanying person to support them, especially when travelling home with their newborn.

Pregnant women living in small communities, especially those residing far from a regional hospital, expressed concerns about having to travel and give birth alone in unfamiliar surroundings. The feeling of loneliness in an unknown community was often linked to the absence of family or close relatives at the place of birth, which could heighten the emotional and psychological challenges of childbirth.

'I don't know this land [the town she is referred to give birth]. I don't have any family, relatives, or friends there. And I don't know what to do when I get there.'

The *Being together or alone* theme highlights the emotional and cultural significance of being surrounded by family during childbirth. Women who gave birth in their home communities emphasised the

value of relational closeness, support, and shared experience. In contrast, women who were referred to regional hospitals often faced childbirth alone, separated from their partners, children, and extended families. This experience was marked by emotional vulnerability, loneliness, and practical concerns, such as travelling home with a newborn without support.

Family formation

All pregnant women expressed the importance of the first meeting between the new baby and its extended family. For them, childbirth was not an isolated event, it was deeply connected to the family as a whole:

> *'The family matters a lot when you are having a new child in the family. There is a need to be together during childbirth. Well, it is everything. Being together as a family in pregnancy and birth are a very important. It is connected.'*

Many of the interviewed women were welcomed at the airport or heliport by family and friends when they returned to their community. The air-or heliports are thereby often the scene of the major and important family event when the family meets a new child for the first time. The centralisation of births thereby transfers the scenes of family life into the public arena and into facilities which are not designed for this kind of event. With the increase of tourism in Kalaallit Nunaat this situation becomes more precarious as an increasing number of persons at the airport or heliport will not be accustomed to this.

For the women who gave birth in their community this event played out very differently. The family usually waits outside the delivery room, typically including grandparents, great-grandparents, siblings, aunts, uncles, cousins, and all others who feel connected to the child. Women, who must travel to give birth, experience, along with their families, the loss of an important part of the birth culture. An event that should establish relationships with the new child within the family and in their community.

> *'Then we have to fly home with our newborn, and that moment of introducing the baby to the family for the first time, it happens at the airport (…) it's not as intimate as it would be if the meeting took place at the hospital [the local hospital]. We've also talked about how it's a bit unfortunate that we even had to discuss this.'*

Separated pregnant women made use of available communication technologies to allow partners, siblings, and other family members to participate in the childbirth experience from a distance. A community member, who has served as a birth companion for women giving birth away from their families, shared her reflections on witnessing childbirth via video call. She described the experience as emotionally ambivalent. On one hand, she appreciated that technology enabled families to be part of such a significant moment. On the other hand, she expressed sadness at seeing partners and siblings trying to connect through a screen during what should be a joyful, shared family event experienced together in the same room, not apart:

> *'It looks tough, fortunately we have good communication options today, but even so, especially for those with other children, if there is no care for the children at home, they can't afford to travel and therefore the father can't be present. That means that the children get the news of their new sibling from a video call. No, it's such a bad feeling to say to the father 'you've become a father to a girl' on a video call.'*

This theme reflects how childbirth, and the first meeting is culturally understood as a moment of family formation and intergenerational continuity. While birth location can disrupt these practices, the underlying values of togetherness, recognition, and relational belonging remain central to how families in Kalaallit Nunaat experience and interpret childbirth.

Identity and sense of belonging

According to the pregnant women in this study, having a child is not only a family event, but also a shared experience in which the entire community is involved. Support from both family members and the wider community played an important role for the interviewed women, offering emotional strength and a sense of belonging throughout the process:

'How should I describe it? For example, the community members here had been very supportive. They told me that everything will be OK, that we are strong together in our family.'

Welcoming a new child into the local community was seen as something that concerns everyone. Pregnant women described feeling supported and cared for by community members—not only during pregnancy but also while giving birth at the regional hospital, even when physically apart. Families and community members stayed connected, waiting together for the good news and celebrating the arrival of the new community member. When the mother and baby returned home healthy, they were warmly welcomed by everyone, marking the beginning of a shared celebration and a new chapter for the whole community.

'When a new child from our community is born, everyone is happy and visits those who have a new child in their family to offer congratulations. When the little child comes home to our community safe and healthy, we welcome it.'

According to pregnant women in small communities, the support of community members played a vital role in fostering a sense of belonging to the community. This support was especially important both emotionally and practically, particularly when returning home after childbirth.

The pregnant women and their families in this study reflected deeply on identity and the sense of belonging in relation to the community where their children will live and grow up. They were highly aware of how being born outside their local community might influence their children's connection to their roots and their developing sense of identity. Many women shared personal reflections on how their own sense of belonging is tied to the place where they were born, emphasising the importance of birthplace in shaping identity and community bonds:

'It's just a bit of a shame to have to say to the child that we parents are from [our local community], and you were born in Nuuk [at national hospital]. I think where you are born gives you your identity. I think place of birth is something about our identity.'

Another family expressed disappointment that their child would not be born in their local community, feeling that this would affect the child's connection to their roots and sense of belonging:

'It was important for us that our child is a maanimioq (one who is from here) [our local community]. It's a shame our child will not be born here where we live.'

Nuna inunngorfik is Kalaallisut for how you speak about place of birth where you were born into your Inuit culture, traditions and community. The pregnant women expressed concern about how their children will identify themselves and develop a sense of belonging when growing up in a community or *nuna* where they were not born.

'Child births in our community gives new energy and happiness to all of us'.

This final theme reflects how childbirth is intimately tied to cultural identity and belonging. While birth location can disrupt traditional practices, the values of community connection, shared celebration, and rootedness in place remain central to how families in Kalaallit Nunaat understand the meaning of birth.

Discussion

Grounded in the Peqqissuserput framework, this qualitative study provides insight into how pregnant women in Kalaallit Nunaat experience childbirth in the context of centralised delivery services, highlighting the profound cultural, emotional, and social implications of giving birth away from one's home community. The findings reveal that centralisation not only affects the logistics of childbirth but also reshapes the cultural landscape surrounding birth, altering how women, families, and communities experience and remember this pivotal life event.

Centralisation and the loss of cultural birth practices

In Kalaallit Nunaat and other Arctic regions, centralisation has led to a significant shift in the birthing experience, particularly for women who must travel to regional hospitals. In 2021, over half (55.4%) of all

births occurred outside the woman's home community [24]. The Referral and Admission Guidelines (2022) specify conditions for local births, including a low-risk pregnancy, a single foetus in head-down position, and spontaneous labour after the 36th week. These women often face childbirth in isolation, separated from their partners, children, and extended families, which leads to emotional stressors [3,4].

This separation disrupts the Inuit birthing culture, where childbirth is a communal event and the woman is the central figure, surrounded by support and care. As one professional described, as part of the larger project the present study was conducted in: "the woman becomes the queen in this undertaking" [5]. The quote reflects not only the honour and attention afforded to the birthing woman but also the broader meaning of community involvement: that women-centred, relational, and culturally embedded care is typically fully available in local settings while in referral hospitals this is challenged due to the absence of family support.

The absence of family during childbirth not only affects the emotional well-being of the mother but also results in a loss of shared memories and collective experiences. Women who give birth away from home often describe a sense of detachment from their community and a diminished sense of identity for their newborns. The concept of *nuna inunngorfik*—the land, place of birth, where you are born into your Inuit culture and traditions—highlights the cultural significance of birthplace in shaping identity and belonging. When children are born outside their home communities, families express concern that their children may lack a vital connection to their roots. This was also described by Fog [25], expressing how the centralisation has not only affected individual pregnant women but also had social consequences for their families, and communities.

Local births therefore matter not only for cultural continuity but for the psychosocial wellbeing of the woman and her family, including the presence of partners, grandparents, and siblings—elements described by participants as central for feeling safe, supported, and grounded. These shared experiences contribute to the family's long-term wellbeing by forming collective memories around the birth, which several participants linked to strengthening family ties and community cohesion.

In addition, to cultural disconnection, centralisation may also limit women's involvement in decisions about their care. As Healy et al [26]. argue, a risk-adverse maternity culture can reduce the autonomy of both midwives and women, shifting decision-making power toward institutional protocols. Barclay et al. [27] similarly advocate for a broader understanding of risk that includes social and cultural dimensions. In Kalaallit Nunaat, where relational well-being and cultural continuity are central, balancing clinical safety with culturally grounded decision-making may help restore a sense of agency and cultural safety in maternity care.

Findings from this study [5], also indicate that women generally wish to give birth in their home communities, while simultaneously expressing a desire for a reliable birth preparedness system. This dual concern reflects the complexity of balancing safety and cultural belonging. Existing literature suggests that cultural disconnection and centralisation may influence decision-making processes for both pregnant women and midwives [4,25,26], though further research is needed to explore these dynamics in depth.

Emotional and practical consequences

The emotional toll of centralised childbirth is evident in the experiences of loneliness, anxiety, and vulnerability reported by women who must travel alone. These feelings are compounded by practical concerns, such as managing travel logistics, financial burdens, and the challenge of returning home with a newborn without support. The study also highlights how centralisation creates social and cultural inequalities, as not all families can afford to be together during childbirth, further marginalising those in remote or economically disadvantaged communities. Such economic stressors were also identified among Canadian Inuit women having to travel for birth [3]. Chamberlain and Barclay [3] discuss how in a Canadian context some may assume that women who prefer local births, where possible, are unaware of medical risks. However, their research suggests that women often weigh those risks against the emotional and cultural importance of being surrounded by family, and that the social disruption of relocation is often experienced as more harmful than the clinical risks involved [3].

In contrast, women who give birth locally describe a sense of calm, support, and continuity. Familiarity with local midwives, the presence of family, and the ability to rest and bond with the newborn in a known

environment contribute to a more positive and culturally grounded birthing experience. Several women emphasised that giving birth in their home community allowed them to focus more fully on the birthing process itself, without the distractions and stressors associated with travel and separation. Additionally, having already received visits from family and friends during their hospital stay created a more peaceful post-partum period, enabling new mothers to return home and rest without the pressure of immediate social obligations. These findings underscore the importance of relational and environmental factors in shaping perceptions of safety and well-being during childbirth.

Implications for birth culture and identity

The findings suggest that centralisation has contributed to the erosion of a culturally embedded birth culture in Kalaallit Nunaat. Birth, a moment of communal celebration and identity formation, is increasingly experienced as a medical event removed from its cultural context. This shift has implications not only for maternal well-being but also for the intergenerational transmission of cultural values and identity.

The loss of shared birthing memories, those created when families gather around the birthing woman, represents a broader cultural loss. These memories are foundational to family narratives and community cohesion. When childbirth occurs away from home, these narratives are fragmented, and the opportunity to collectively welcome a new member into the community is diminished.

Similar challenges have been documented among Indigenous communities in Canada, where centralised maternity care has prompted community-based efforts to reclaim culturally relevant birthing practices [28,29]. Initiatives such as the *Inuulitsivik Midwifery Service* in Nunavik [30] and the "Bring Birth Home" movement in Manitoba [31] demonstrate the potential for community-led models that integrate Indigenous knowledges with biomedical care. These models emphasise the importance of local births for cultural continuity and maternal well-being. They also offer practical examples of how culturally grounded, community-led approaches can be successfully integrated into formal healthcare systems, insights that may inform future policy development in Kalaallit Nunaat.

Strengths and limitations

A key strength of this study is its grounding in the Peqqissuserput framework and the analysis being led by an Indigenous researcher, ensuring that the analysis reflects Kalaallit epistemologies [32]. Additionally, the inclusion of participants from all five health regions provides a broad geographic perspective, enhancing the relevance of the findings across diverse community contexts. Furthermore, the participants' varied experiences with local births and referral-based births provided a nuanced understanding of how place and mobility influence birthing experiences in Kalaallit Nunaat. In addition, the elderly participants contributed long-term, community-rooted insights into birthing practices and the role of local childbirth in earlier generations. The interviews were conducted, translated, and analysed by the same researcher, which allows for a deeper, more nuanced understanding pf each participant's narrative [18]. A limitation of the study is that it does not include the perspectives of women who have experienced complicated births, which may have influenced their views on centralised services, risk and cultural safety in different ways.

Conclusion

This study highlights how centralised delivery services in Kalaallit Nunaat affect not only the logistics of maternity care but also the cultural, emotional, and social dimensions of the birthing experience. Findings that align with studies from other Arctic regions. For many pregnant women, being referred away from their home communities results in emotional strain, loss of familial support, and a diminished sense of identity and belonging for both mother and child. The findings underscore that childbirth is not solely a clinical event but a deeply relational and cultural process, embedded in family, community, memory, and identity and connected to place.

By drawing on the Peqqissuserput framework and the voices of pregnant women and their families across all five health regions, this study reveals the importance of culturally grounded care that respects Inuit values and supports local birthing options where possible. The experiences shared by participants

point to the need for a more balanced approach to maternity care, one that is both clinically sound and culturally responsive.

Community-led models from other Indigenous contexts offer promising pathways for reimagining maternity care in Kalaallit Nunaat, where cultural continuity, family presence, and shared decision-making are central to well-being. Compared to other Arctic regions, births in Kalaallit Nunaat still take place within the country, often with Kalaallit Inuit midwives and in settings where Kalaallisut can be spoken. There is potential to foster culturally sensitive birthing experiences. However, it remains essential to recognise and address the logistical and cultural complexities that pregnant families face when travelling to give birth. Listening to and learning from these families is key to improving care and ensuring that cultural sensitivity is strengthened, even when birth cannot take place locally.

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Author contributions

CRediT: **Ingelise Olesen:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – review & editing; **Christine Ingemann:** Writing – original draft, Writing – review & editing; **Christina Viskum Lytken Larsen:** Conceptualization, Supervision, Writing – review & editing; **Charlotte Brandstrup Ottendahl:** Formal analysis, Investigation, Methodology, Writing – review & editing; **Sofie Emma Rubin:** Investigation, Writing – review & editing; **Peter Bjerregaard:** Supervision, Writing – review & editing; **Tenna Jensen:** Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – review & editing.

Disclosure statement

The authors declare that they have no competing interest. An AI language model (Microsoft Copilot 365) was used to assist with wording improvements in parts of the manuscript. All content was reviewed and edited by the authors.

ORCID

Ingelise Olesen 0009-0007-8717-0801
Christine Ingemann 0000-0002-1390-4952
Christina Viskum Lytken Larsen 0000-0002-6245-4222
Charlotte Brandstrup Ottendahl 0000-0002-9285-3099
Tenna Jensen 0000-0001-9020-0089

Data availability statement

The datasets analysed in the study are not publicly available due to the difficulty to de-identify qualitative data in a small population like Kalaallit Nunaat, therefore data are not available.

Ethics approval and consent to participate

The study has been performed in accordance with the Declaration of Helsinki and the Greenlandic Science Ethics Committee (Danish: Videnskabsetisk Udvalg) granted ethical approval of the project. Each participant was provided with an informed consent form in either Danish or Greenlandic, and the content was explained. Participants signed the form to document their informed agreement and received a copy with contact details.

Authors' information

The corresponding and first author IO is an Indigenous researcher. The remaining authors are non-indigenous researchers.

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